

Class #17 - Vector spaces and Subspaces

1. Def: A (*real*) *vector space* is a _____ set V of objects, called _____, on which are defined two operations, called _____ and _____ by scalars (real numbers), subject to the ten axioms (or rules) listed below. The axioms must hold for *all* vectors u, v , and w in V and for all scalars c and d .

(1) The sum of u and v , denoted by _____, is in _____.

(2) $u + v =$ _____

(3) $(u + v) + w =$ _____

(4) There is a *zero* vector 0 in _____ such that $u + 0 =$ _____ for _____ u in _____.

(5) For each u in V , there is a vector $-u$ such that $u + (-u) =$ _____

(6) The scalar multiple of u by c , denoted by _____, is in V .

(7) $c(u + v) =$ _____

(8) $(c + d)u =$ _____

(9) $c(du) =$ _____

(10) $1u = u$

2. Examples:

(a) $\mathbf{R}^n$

(b) $\mathbf{R}^{m \times n}$

(c) $\mathbf{P}_n := \{p(t) = a_0 + a_1t + \cdots + a_nt^n : a_i \in \mathbf{R}\}$

(d) The set of all continuous functions defined on a set I .

(e) How about the set of all differentiable functions defined on a set I ??

(f) The set of all infinite sequence of real numbers = $\{(a_0, a_1, a_2, a_3, \cdots, a_n, \cdots) : a_i \in \mathbf{R}\}$

3. The following facts can be shown using 10 axioms above:

(a) The zero vector, 0 , is unique:

Proof: Suppose there is $w \in V$ such that $u + w = u$ for _____ $u \in V$.

Then _____ $+ w = 0$ because _____ in V . Moreover, $0 + w =$ _____ by axiom _____.

Hence, $0 =$ _____ $+ w =$ _____.

(b) The negative of u , $-u$, is the unique vector in V such that $u + (-u) = 0$.

Proof: Suppose there is $w \in V$ such that _____.

Then, _____ $= -u +$ _____ $= -u + (u + w) =$ _____ $+ w = w$.

(c) $0u = 0$ for every $u \in V$.

Proof: $0u = (\text{_____} + \text{_____})u = \text{_____} + \text{_____}$

Then, $0 = (-0u) + 0u = (-0u) + \text{_____} + \text{_____} = \text{_____} + 0u = 0 + 0u =$ _____. Hence, $0 = 0u$.

(d) $c0 = 0$ for every scalar c :

Proof: $c0 = c(0 + 0) = \text{_____} + \text{_____}$. Then,

$0 = -c0 + c0 = (-c0 + c0) + \text{_____} = 0 + c0 = \text{_____}$.

(e) $(-1)u = -u$ for all $u \in V$:

Proof: $u + (-1)u = \text{_____}u + (-1)u = (\text{_____} + \text{_____})u = 0u = \text{_____}$.

Hence, $(-1)u = -u$.

4. Def: A *subspace* H is a _____ of a vector space V which itself is a vector space.

Example: $V = C([0, 1])$ and $H = \mathbf{P}_n$

5. Note: $H \subseteq V$ of a vector space V is a _____ of V if

(a) The zero vector of _____ is in _____.

(b) H is closed under vector addition, that is, for each u and v in _____, $u + v \in$ _____.

(c) H is closed under scalar multiplication, that is, for each u in _____, and a scalar c , the vector _____ is in _____.

6. Examples:

- (a) V any vector space, and $H = \{0\}$.

Such H is called the *zero subspace*.

- (b) $V = \mathbf{R}^3$. $H = \mathbf{R}^2$.

Is H a subspace of V ??

- (c) V any vector space, and $H = \text{Span}\{v_1, v_2\}$ where $v_1, v_2 \in V$.

Theorem: If $v_1, \dots, v_p$ are in a _____ space V , then $H =$ _____ is a subspace of _____.

We call H the *subspace spanned* or (*generated*) by $\{v_1, \dots, v_p\}$.

Def: Given any subspace H of a _____ space V , a *spanning* (or *generating*) *set* for H is a set $\{v_1, \dots, v_p\}$ in _____ such that $H =$ _____.

Example: Let $V = C([0, 1])$ and $H = \mathbf{P}_n$. Give a generating set for H .

(d) Given $A \in \mathbf{R}^{m \times n}$, the set of all solutions to the homogeneous equation $Ax = 0$ is a subspace of _____: